

MATH 151: ALGEBRA

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Algebra of Complex Numbers

We find that the set of Reals, $\mathbf{R}$ is not closed under the extraction of some roots. In particular, the problem:

$$x^2 = -1 \quad (1)$$

has no solution in $\mathbf{R}$.

Hence we enlarge $\mathbf{R}$, so that the square roots of negative numbers can be extracted. Thus we look for an enlarged set, $\mathbf{C}$, called the Complex numbers.

Cont...

Complex numbers arise most directly out of solving polynomial equations.

Consider the quadratic equation

$$z^2 - 4z + 5 = 0 \quad (2)$$



Cont...

The equation above has two solutions z_1 and z_2 such that

$$(z - z_1)(z - z_2) = 0 \quad (3)$$

Using the familiar formula for the roots of a quadratic equation, the solutions z_1 and z_2 written in brief as

$$z_{1,2} = \frac{4 \pm \sqrt{(-4)^2 - 4(1 \times 5)}}{2} = 2 \pm \frac{\sqrt{-4}}{2} \quad (4)$$

Cont...

The second term on the RHS of (4) is called an **imaginary term** since it contains the square root of a negative number, the first term is called a **real term**. The full solution is the sum of a real term and an imaginary term and it is called a **Complex Number**.



Notation

The choice of the symbol z for the quadratic variable was not arbitrary; the conventional representation of a complex number is z , where z is the sum of a real part x and i times an imaginary part y , i.e.

$$z = x + iy, \tag{5}$$

where i is used to denote the square root of -1 .

Cont...

The real part x and the imaginary part y are usually denoted by $\text{Re}(z)$ and $\text{Im}(z)$ respectively.

Engineers, in particular, use j instead of i . However, for consistency, we will use i .



Cont...

In our particular example $\sqrt{-4} = 2\sqrt{-1} = 2i$,
and hence the two solutions of (3) are

$$z_{1,2} = 2 \pm \frac{2i}{2} = 2 \pm i \quad (6)$$

Thus, here $x = 2$ and $y = \pm 1$.



Cont...

For compactness, a complex number is sometimes written in the form

$$z = (x, y), \quad (7)$$

where the components z may be thought of as coordinates in an xy -plot.



Addition And Subtraction of Complex Numbers

- The addition of two complex numbers z_1 and z_2 , in general, gives another complex number.
- The real and imaginary parts add similarly to real numbers:

$$z_1 + z_2 = (x_1 + y_1 i) + (x_2 + y_2 i) = (x_1 + x_2) + (y_1 + y_2) i \quad (8)$$



Cont...

Or in component notation:

$$z_1 + z_2 = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \quad (9)$$

By straightforward application of commutativity and associativity, i.e.,

$$z_1 + z_2 = z_2 + z_1$$

$$z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3$$



Example 1

Question:

Sum the complex numbers $1 + 2i, 3 - 4i, -2 + i$.

Solution:

Summing the real terms: $1 + 3 - 2 = 2$,

Summing the imaginary terms: $2i - 4i + i = -i$,

Hence, $(1 + 2i) + (3 - 4i) + (-2 + i) = 2 - i$.

The subtraction of complex numbers is very similar to their addition.



Equality

Let $z_1 = x_1 + y_1i$ and $z_2 = x_2 + y_2i$ be complex numbers, then:

$$z_1 = z_2$$

if and only if

$$x_1 = x_2 \quad \text{and} \quad y_1 = y_2$$



Modulus And Argument

The **modulus** of the complex number z is denoted by $|z|$ and is defined as

$$|z| = \sqrt{x^2 + y^2}. \quad (10)$$

Hence the modulus of the complex number is the distance of the corresponding point from the origin in the Argand diagram.



Cont...

The **argument** of the complex number z is denated by **arg z** and is defined as

$$\text{arg } z = \tan^{-1} \left(\frac{y}{x} \right) \quad (11)$$

It can be seen that $\arg z$ is the angle that the line joining the origin to z on the Argand diagram makes with the positive x -axis. The anticlockwise direction is taken to be positive by convention.



Cont...

Account must be taken of the signs of x and y individually in determining in which quadrant $\arg z$ lies.



Example 2

Question

Find the modulus and the argument of the complex number $z = 2 - 3i$.



Cont...

Solution

The modulus is given by

$$|z| = \sqrt{(2)^2 + (-3)^2} = \sqrt{13}$$

the argument is given by

$$\arg z = \theta = \tan^{-1} \left(\frac{-3}{2} \right) = -56.31$$



Cont...

Since $x = 2$ and $y = -3$, z clearly lies in the fourth quadrant; (i.e., the argument must be measured from the positive real axis), therefore $\arg z = 56.31^\circ$ is the appropriate answer.



Multiplication

The product of two complex numbers z_1 and z_2 is found by multiplying them out in full and remembering that $i^2 = -1$.

$$z_1 z_2 = (x_1 + y_1 i)(x_2 + y_2 i) \quad (12)$$

$$= x_1 x_2 + x_1 y_2 i + y_1 x_2 i + y_1 y_2 i^2 \quad (13)$$

$$= (x_1 x_2 - y_1 y_2) + (x_1 y_2 + y_1 x_2) i \quad (14)$$



Example 3

Question

Multiply the complex numbers $z_1 = 3 + 2i$ and $z_2 = -1 - 4i$.

Solution

By direct multiplication , we find

$$z_1 z_2 = (3 + 2i)(-1 - 4i) = -3 - 2i - 12i - 8i^2 = 5 - 14i$$



Cont...

The multiplication of complex numbers is both commutative and associate. i.e

$$z_1 z_2 = z_2 z_1 \quad (15)$$

$$(z_1 z_2) z_3 = z_1 (z_2 z_3) \quad (16)$$

$$|z_1 z_2| = |z_1| |z_2| \quad (17)$$

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2. \quad (18)$$



Example 4

Question

Verify that $|z_1 z_2| = |z_1| |z_2|$, holds for the product of $z_1 = 3 + 2i$ and $z_2 = -1 - 4i$



Cont...

Solution

$$|z_1 z_2| = |5 - 14i| = \sqrt{5^2 + (-14)^2} = \sqrt{221}$$

We also find

$$|z_1| = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$|z_2| = \sqrt{(-1)^2 + (-4)^2} = \sqrt{17}$$

and hence

$$|z_1||z_2| = \sqrt{13}\sqrt{17} = \sqrt{221} = |z_1 z_2|$$

Complex Conjugate

If z has the convenient form $x + yi$ then the **complex conjugate**, denoted by $\bar{z}$ may be found simply by **changing the sign of the imaginary part**, i.e, if

$$z = x + yi$$

then

$$\bar{z} = x - yi$$



Example 5

Question

Find the complex conjugate of

$$z = a + 2i + 3ib$$

Solution:

$$\bar{z} = a - (2 + 3b)i$$



Example 6

Question

Find the complex conjugate of the complex number $z = w^{(3y+2ix)}$ where $w = x + 5i$.



Cont...

Solution

In this case , w itself contains real and imaginary components and so must be written out in full, i.e.

$$z = w^{(3y+2ix)} = (x + 5i)^{(3y+2ix)}$$

Now we can replace each i by $-i$ to obtain

$$\bar{z} = (x - 5i)^{(3y-2ix)}$$

It can be shown that the product $z\bar{z}$ is real as required.

Properties

The following properties of complex conjugate are easily proved and others maybe derived from them. If $z = x + yi$ then

(a) $\bar{\bar{z}} = z$

(b) $z + \bar{z} = 2 \operatorname{Re} z = 2x$

(c) $z - \bar{z} = 2 \operatorname{Im} z = 2yi$



Cont...

(d) $\frac{z}{\bar{z}} = \left(\frac{x^2 - y^2}{x^2 + y^2} \right) + i \left(\frac{2xy}{x^2 + y^2} \right)$

(e) $\overline{z_1 z_2} = \overline{z_2} \overline{z_1}$ for two complex numbers z_1 and z_2 .

(f) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ for two complex numbers z_1 and z_2 .



Division

The division of two complex numbers z_1 and z_2 bears some similarity to their multiplication.

Writing the quotient in the component form we obtain

$$\frac{z_1}{z_2} = \frac{x_1 + y_1 i}{x_2 + y_2 i} \quad (19)$$



Cont...

In order to separate the real and the imaginary components of the quotient, we multiply both numerator by the complex conjugate of the denominator.

By definition , this process will leave the denominator as a real quantity.



Cont...

$$\frac{z_1}{z_2} = \frac{(x_1 + y_1 i)(x_2 + y_2 i)}{(x_2 + 2i)(x_2 - y_2 i)} = \frac{(x_1 x_2 + y_1 y_2) + (x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2} \quad (20)$$

$$= \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + i \frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2} \quad (21)$$

Hence we have separated quotient into real and imaginary components as required.



Example 7

Question

Express z in the form $x + iy$, when $z = \frac{3 - 2i}{-1 + 4i}$



Cont...

Solution

Multiplying numerator and denominator by the complex conjugate of the denominator we obtain

$$z = \frac{(3 - 2i)(-1 - 4i)}{(-1 + 4i)(-1 - 4i)} = \frac{-11 - 10i}{17} = \frac{-11}{17} - \frac{10i}{17}$$



Properties

$$(a) \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

$$(b) \arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$$



Exercise 1

1. Given $z_1 = 2 + 4i$, $z_2 = 3 - i$ and $z_3 = -3i$ Determine:

(a) $z_1 + z_2$

(b) $z_1 - z_2$

(c) $z_1 - z_3$

2. Find

(a) $(1 - 8i)(1 + 8i)$

(b) $(4 + i)(3i)$

(c) $(2 - i)(5 + 2i)$



Cont...

3. Write each in standard form.

(a) $\frac{3-i}{2+\frac{7i}{3}}$

(b) $\frac{9-i}{8i}$

(c) $\frac{1+2i}{1+2i}$



Polar Representation Of Numbers

The polar representation of a complex number is easier to manipulate. The polar representation makes use of the **complex exponential function** which is defined by

$$e^z = \exp z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \quad (22)$$

Note that we may have from (22)

$$e^{z_1} e^{z_2} = e^{z_1 + z_2} \quad (23)$$



Cont...

From (22), it immediately follows that for $z = i\theta$,

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \dots \quad (24)$$

$$= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots + i\theta - i\frac{\theta^3}{3!} + i\frac{\theta^5}{5!} - \dots \quad (25)$$

which is a sum of series expansion of $\cos\theta$ and $\sin\theta$



Cont...

and hence we obtain

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (26)$$

called the **Euler's equations**

It also follows from (26)

$$e^{in\theta} = \cos n\theta + i \sin n\theta \quad (27)$$

for all n .



Cont...

From Euler's equation (26) we deduce that

$$r e^{i\theta} = r(\cos \theta + i \sin \theta) \quad (28)$$

$$= x + yi \quad (29)$$

Thus a complex number may be represented in the polar form

$$z = r e^{i\theta} \quad (30)$$



Cont...

Now we can identify r with $|z|$ and θ with $\arg z$. The angle θ lies conventionally in the range $-\pi < \theta \leq \pi$, but , since rotation by θ is the same as rotation by $2n\pi + \theta$, where n is any integer,

$$re^{i\theta} \equiv re^{i(\theta+2n\pi)} \quad (31)$$



Cont...

The algebra for the polar representation is different from that of the real and imaginary component representation, though, of course, the results are identical.

Similarly from the Cartesian plane, the polar to Cartesian co-ordinate transformation equation can be written as follows



Cont...

$$x = r \cos \theta \quad y = r \sin \theta \quad (32)$$

where $r = |z|$, the modulus of z , and $\theta = \tan^{-1} \left(\frac{y}{x} \right)$, the argument of z . Thus, a complex number in polar form is given by

$$z = x + iy \quad (33)$$

$$= r \cos \theta + i r \sin \theta \quad (34)$$

$$= r(\cos \theta + i \sin \theta) \quad (35)$$



Example 8

Question

Express $z = 3 + 4i$ in polar form.

Solution

Since $|z| = 5$ and $\arg(z) = 53.13^\circ$, hence we may write the polar form of z explicitly as

$$z = 5 \cos 53.13^\circ + i \sin 53.13^\circ$$



MULTIPLICATION AND DIVISION IN POLAR FORM

Multiplication and division in polar form are particularly simple. The product of $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ is given by

$$z_1 z_2 = r_1 e^{i\theta_1} r_2 e^{i\theta_2} \quad (36)$$

$$= r_1 r_2 e^{(\theta_1 + \theta_2)} \quad (37)$$

$$= r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)] \quad (38)$$



Cont...

Division is equally simple in polar form; the quotient of z_1 and z_2 is given by

$$\frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} \quad (39)$$

$$= \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)] \quad (40)$$



De MOIVRE'S THEOREM

We now derive an extremely important theorem. Since

$$(e^{i\theta})^n = e^{in\theta} \quad (41)$$

we have

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad (42)$$

where the identity

$$e^{in\theta} = \cos n\theta + i \sin n\theta \quad (43)$$

follows from the series definition of $e^{in\theta}$.



Cont...

The result is called **De Moivre's theorem**. The theorem is valid for all n whether real imaginary or complex.

The theorem will be applied to

- proves of trigonometric identities,
- finding the n th roots of unity, and
- solving polynomial equations with complex roots.



TRIGONOMETRIC IDENTITIES

The use of **De Moivre's theorem** in finding trigonometric identities is best illustrated by examples. We consider the expression of a multiple - angle function in terms of a polynomial in the single-angle function, and its converse.



Example 9

Question

Express $\sin 3\theta$ and $\cos 3\theta$ in terms of powers of $\cos \theta$ and $\sin \theta$.



Cont...

Solution

Using de Moivre's theorem

$$\begin{aligned}\cos 3\theta + i \sin 3\theta &= (\cos \theta + i \sin \theta)^3 \\ &= (\cos^3 \theta - 3 \cos \theta \sin^2 \theta) + i(3 \sin \theta \cos^2 \theta - \sin^3 \theta)\end{aligned}$$



Cont...

Solution

We can equate the real and imaginary part coefficients separately, i.e,

$$\begin{aligned}\cos 3\theta &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\ &= 4 \cos^3 \theta - 3 \cos \theta\end{aligned}$$



Cont...

Solution

$$\begin{aligned}\sin 3\theta &= 3\sin\theta\cos^2\theta - \sin^3\theta \\ &= 3\sin\theta - 4\sin^3\theta\end{aligned}$$

This method can clearly be applied to finding power expansions of $\cos n\theta$ for any positive integer n .



NOTE

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad (44)$$

$$z^n - \frac{1}{z^n} = 2i \sin \theta \quad (45)$$

These equalities follow from simple applications of de Moivre's theorem



Cont...

i.e.

$$z_n + \frac{1}{z_n} = (\cos \theta + i \sin \theta)^n + (\cos \theta + i \sin \theta)^{-n} \quad (46)$$

$$= \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta) \quad (47)$$

$$= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta = 2 \cos n\theta$$



Cont...

and

$$z^n - \frac{1}{z^n} = (\cos \theta + i \sin \theta)^n - (\cos \theta + i \sin \theta)^{-n} \quad (48)$$

$$= \cos n\theta + i \sin n\theta - \cos(-n\theta) + i \sin(n\theta) \quad (49)$$

$$= \cos n\theta + i \sin n\theta - \cos n\theta + i \sin n\theta = 2i \sin n\theta$$



Cont...

In particular case where $n = 1$

We have,

$$z + \frac{1}{z} = e^{i\theta} + e^{-i\theta} = 2\cos\theta \quad (50)$$

$$z - \frac{1}{z} = e^{i\theta} - e^{-i\theta} = 2i\sin\theta \quad (51)$$



Example 10

Question

Find an expression for $\cos^3 \theta$ in terms of $\cos 3\theta$ and $\cos \theta$

Solution

$$\cos^3 \theta = \frac{1}{2^3} \left(z + \frac{1}{z} \right)^3 \quad (52)$$

$$= \frac{1}{8} \left(z^3 + 3z + \frac{3}{z} + \frac{1}{z^3} \right) \quad (53)$$

$$(54)$$

Cont...

Solution

$$= \frac{1}{8}\left(z^3 + \frac{1}{z^3}\right) + \frac{3}{8}\left(z + \frac{1}{z}\right) \quad (55)$$

we find that

$$\cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta \quad (56)$$



Exercise 2

Using De Moivre's theorem

1. Expand

(a) $\cos 4\theta$ and $\sin 4\theta$

(b) $\cos^3 \theta$ and $\sin^3 \theta$

2. Show that

(a) $\cos 3\theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta$

(b) $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$

(c) $\tan 3\theta = \frac{3\tan \theta - \tan^3 \theta}{1 - 3\tan^2 \theta}$



Finding The Nth Roots Of Unity

The equation $z^2 = 1$ has the familiar solutions $z = \pm 1$. In complex numbers, we can solve the general equation $z^n = 1$. And we know that the equation has n solutions. In order to proceed, we rewrite the equation as

$$z^n = e^{2ik\pi} \quad (57)$$

where k is any integer.



Cont...

Now taking the n th root of each side of the equation, we find

$$z = e^{\frac{2ik\pi}{n}} \quad (58)$$

Hence, the solutions of $z^n = 1$ are

$$z_{1,2,\dots,n} = 1, e^{\frac{2i\pi}{n}}, \dots, e^{\frac{2i(n-1)\pi}{n}}, \quad (59)$$

corresponding to the values $k = 0, 1, 2, \dots, n-1$



NOTE

Larger integer values of k do not give new solutions, since the roots already listed are simply cyclically repeated for $k = n, n + 1, n + 2$, etc.



Example 11

Question

Find the solutions of the equation $z^3 = 1$.

Solution

By applying the above method we find

$$z = e^{\frac{2ik\pi}{3}}$$

Hence the three solutions are $z_1 = e^{0i} = 1, z_2 = e^{\frac{2i\pi}{3}}, z_3 = e^{\frac{4i\pi}{3}}$.

Cont...

Solution

We note that, as expected, the next solution for which $k = 3$, gives $z_4 = e^{\frac{6i\pi}{3}} = 1 = z_1$, so that there are only three separate solutions.



Solving Polynomial Equations

A third application of de Moivre's theorem is the **solution of polynomial equations**. Complex equations in the form of a polynomial relationship must first be solved for z in a similar fashion to the method for finding the n roots of real polynomial equations. Then the complex roots of z may be found.



Example 12

Question

Solve the equation $z^6 - z^5 + 4z^4 - 6z^3 + 2z^2 - 8z + 8 = 0$

Solution

We first factorize to give

$$(z^3 - 2)(z^2 + 4)(z - 1) = 0$$

Hence $z^3 = 2$ or $z^2 = -4$ or $z = 1$. The solutions to the quadratic equation are $z = \pm 2i$;

Cont...

Solution

to find the complex cube roots, we first write the equation in the form

$$z^3 = 2 = 2e^{2ik\pi}$$

where k is any integer. If we now take the cube root, we get

$$z = 2^{\frac{1}{3}} e^{\frac{2ik\pi}{3}}$$

Cont...

Solution

To avoid the duplication of solutions, we use the fact that $-\pi < \arg z \leq \pi$ and find

$$\begin{aligned} z_1 &= 2^{\frac{1}{3}} \\ z_2 &= 2^{\frac{1}{3}} e^{\frac{2\pi i}{3}} = 2^{\frac{1}{3}} \left(-\frac{1}{2} + \frac{\sqrt{3}}{2} i \right) \\ z_3 &= 2^{\frac{1}{3}} e^{-\frac{2\pi i}{3}} = 2^{\frac{1}{3}} \left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right) \end{aligned}$$

Complex Logarithm And Complex Powers

We can define the logarithm of a complex number and we can use complex numbers as exponents. Let us denote the natural logarithm of a complex number z by

$$\omega = \ln z,$$

where the notation $\ln$ will be explained shortly. Thus, ω must satisfy

$$z = e^{\omega} \tag{60}$$



Cont...

we see that

$$z_1 z_2 = e^{\omega_1} e^{\omega_2} = e^{\omega_1 + \omega_2}$$

and taking logarithms of both sides we find

$$\ln(z_1 z_2) = \omega_1 + \omega_2 = \ln z_1 + \ln z_2 \quad (61)$$



Cont...

We may investigate further the properties of $\ln z$. We have already noted that the argument of a complex number is multivalued, i.e.

$$\arg z = \theta + 2n\pi$$

where n is any integer. Thus, in polar form, the complex number z should strictly be written as

$$z = r e^{i(\theta + 2n\pi)}$$

(62) 

Cont...

Taking the logarithm of both sides and using (28), we find

$$\ln z = \ln r + i(\theta + 2n\pi) \quad (63)$$

where $\ln r$ is the natural logarithm of the real positive quantity r and so is written normally. Thus from (63) we see that $\ln z$ is itself multivalued.



Cont...

To avoid this multivalued behaviour it is conventional to define another function $\ln z$, the principal value of $\ln z$, which is obtained from $\ln z$ by restricting the argument of z to lie in the range $-\pi \leq \theta \leq \pi$.



Example 13

Question

Evaluate $\ln(-i)$

Solution

By rewriting i as a complex exponential , we find

$$\ln(-i) = \ln \left[e^{i(-\pi/2 + 2n\pi)} \right] = i(-\pi/2 + 2n\pi) \quad (64)$$

where n is any integer.



Cont...

Solution

Hence

$$\ln(-i) = -\frac{\pi}{2}, \frac{3i\pi}{2} \quad (65)$$

We note that, the principal value of $\ln(-i)$, is given by

$$\ln(-i) = -\frac{\pi}{2} \quad (66)$$



Complex Exponential

Example

Simplify the expression $z = i^{-2i}$

Solution

First we take the logarithm of both sides of the equation:

$$\ln z = -2i \ln i \quad (67)$$



Cont...

Solution

Now inverting the process we find

$$e^{\ln z} = z = e^{-2i \ln i} \quad (68)$$

We can write $i = e^{i(-\pi/2+2n\pi)}$, where n is any integer, and hence

$$\begin{aligned} \ln i &= \ln \left[e^{i(\pi/2+2n\pi)} \right] \\ &= i(\pi/2 + 2n\pi) \end{aligned}$$

Cont...

Solution

We can now simplify z to give

$$\begin{aligned} i^{-2i} &= e^{-2i \times i(\frac{\pi}{2} + 2n\pi)} \\ &= e^{\pi + 4n\pi} \end{aligned}$$

which is a real quantity rather than a complex one.



Exercise 3

Evaluate

① $\ln(2 - 3i)$

② $\ln(-2 + i)$

③ 2^{1+i}

④ $(2 - i)^i$



Vector Algebra

Definition

The term **vector** is used by scientists to indicate a quantity that has both magnitude and direction.

Examples: displacement or velocity or force.

A vector is represented by an arrow or a directed line segment ($\longrightarrow$). The length of the arrow represents **magnitude** of the vector and the arrow points in the **direction of the vector**.



Cont...

We denote a vector by the letter **u**.

A vector $\overrightarrow{AB}$ has initial point **A** (the tail) and terminal point **B** (the tip). We say that **u** and **v** are equivalent (or equal) and we write **u = v**. The zero vector denoted by 0 , has length 0 . **It is the only vector with no specific direction.**



Combining Vectors

Suppose a particle moves from A to B , so its displacement vector is $\overrightarrow{AB}$. The particle changes direction and moves from B to C , with displacement vector $\overrightarrow{BC}$.

The combined effect of these displacements is that the particle has moved from A to C .



Cont...

The resulting displacement vector $\overrightarrow{AC}$ is called the sum of $\overrightarrow{AB}$ and $\overrightarrow{BC}$ written as;

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$

In general , if we start with vectors $\mathbf{u}$ and $\mathbf{v}$, we first move $\mathbf{v}$ so that its tail coincides with the tip of $\mathbf{u}$ and define the sum of $\mathbf{u}$ and $\mathbf{v}$ as follows.



Definition Of Vector Addition

1. If $\mathbf{u}$ and $\mathbf{v}$ are vectors positioned so the initial point of $\mathbf{v}$ is at the terminal point of $\mathbf{u}$,
2. then the sum $\mathbf{u} + \mathbf{v}$ is the vector from the initial point of $\mathbf{u}$ to the final point $\mathbf{v}$.
3. If we place $\mathbf{u}$ and $\mathbf{v}$ so they start at the same point, then $\mathbf{u} + \mathbf{v}$ lies along the diagonal of the parallelogram with $\mathbf{u}$ and $\mathbf{v}$ as sides. (Parallelogram Law.)



Scalar Multiplication

4. If c is a scalar and $\mathbf{u}$ is a vector then the scalar multiple $c\mathbf{u}$ is the vector
- (a) whose length is $|c|$ times the length of $\mathbf{u}$ and
 - (b) whose direction is the same as $\mathbf{u}$ if $c > 0$ and
 - (c) whose direction is opposite to $\mathbf{u}$ if $c < 0$.



Cont...

5. If $c = 0$ or $\mathbf{u} = 0$, then $c\mathbf{v} = 0$.
6. Two non-zero vectors are parallel if they are scalar multiples of one another.
7. The vector $-\mathbf{u} = (-1)\mathbf{u}$ has the same length as $\mathbf{u}$ but points in the opposite direction of $\mathbf{u}$. We call it the negative of $\mathbf{u}$. By the difference $\mathbf{u} - \mathbf{v}$ of two vectors we mean

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$$



Parallel Vectors

Two vectors are **parallel** if they have the **same direction** or are in **exactly opposite directions**.

Suppose that a and b are parallel vectors. If they are parallel there must be a number c so that ,

$$a = cb$$

So, two vectors are parallel if one is a scalar multiple of the other.



Components Of A Vector

If we place the initial point of a vector a at the origin of a rectangular coordinates system, then the terminal point of a has coordinates of the form (a_1, a_2) or (a_1, a_2, a_3) , depending on whether our coordinate system is two or three -dimensional. These coordinates are called the **components of the vector** and we write

$$a = \langle a_1, a_2 \rangle \text{ or } a = \langle a_1, a_2, a_3 \rangle$$



Cont...

We use the notation $a = \langle a_1, a_2 \rangle$ for the ordered pair that refers to a vector so as not to confuse it with the ordered pair (a_1, a_2) that refers to a point in the plane.

In three dimensions, the vector $a = \overrightarrow{OP} = \langle a_1, a_2, a_3 \rangle$ is the position vector of the point $P(a_1, a_2, a_3)$.

Let's consider any other representation $\overrightarrow{AB}$ of a , where the initial point is $A(x_1, y_1, z_1)$ and the terminal point is $B(x_2, y_2, z_2)$.



Cont...

Then we must have

$$x_1 + a_1 = x_2, y_1 + a_2 = y_2 \text{ and } z_1 + a_3 = z_3$$

and so

$$a_1 = x_2 - x_1, a_2 = y_2 - y_1, \text{ and } a_3 = z_2 - z_1$$

Given the points $A(x_1, x_2, x_3)$ and $B(x_2, y_2, z_2)$, the vector with representation $\overrightarrow{AB}$ is

$$\overrightarrow{AB} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$



Example 14

Question

Find the vector represented by the directed line segment with initial point $A(1, -3, 4)$ and terminal point $B(-2, 1 - 1)$.

Solution

the vector corresponding to $\overrightarrow{AB}$ is

$$a = \langle -2 - 1, 1 - (-3), 1 - 4 \rangle = \langle -3, 4, -3 \rangle$$



Cont...

Magnitude

The magnitude or length of the vector is the length of any of its representations and is denoted by the symbol $|v|$. By using the distance formula to compute the length of a segment OP , we obtain the following formulas.



Cont...

The length of the two-dimensional vector $a = \langle a_1, a_2 \rangle$ is

$$|a| = \sqrt{a_1^2 + a_2^2}$$

The length of the three-dimensional vector $\langle a_1, a_2, a_3 \rangle$ is

$$|a| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$



Cont...

If $a = \langle a_1, a_2 \rangle$ and $b = \langle b_1, b_2 \rangle$, then

$$a + b = \langle a_1 + b_1, a_2 + b_2 \rangle$$

$$a - b = \langle a_1 - b_1, a_2 - b_2 \rangle$$

$$ca = \langle ca_1, ca_2 \rangle$$

Similarly, for three-dimensional vectors,



Cont...

$$\langle a_1, a_2, a_3 \rangle + \langle b_1, b_2, b_3 \rangle = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

$$\langle a_1, a_2, a_3 \rangle - \langle b_1, b_2, b_3 \rangle = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

$$c\langle a_1, a_2, a_3 \rangle = \langle ca_1, ca_2, ca_3 \rangle$$



Properties of Vectors

If a, b and c are vectors in V_n and c and d are scalars, then

(a) $a + b = b + a$

(e) $a + 0 = 0$

(b) $a + (b + c) = (a + b) + c$

(f) $(c + d)a = ca + da$

(c) $a + (-a) = 0$

(g) $(cd)a = c(da)$

(d) $c(a + b) = ca + cb$

(h) $1a = a$



The Standard Basis Vectors i, j and k

Three vectors in V_3 play a special role. Let

$$i = \langle 1, 0, 0 \rangle \quad j = \langle 0, 1, 0 \rangle \quad k = \langle 0, 0, 1 \rangle$$

These vectors i, j and k are called the **standard basis vectors**. They have **length 1** and **point in the directions** of the positive x, y and z – *axes*. Similarly, in two dimensions we define $i = \langle 1, 0 \rangle$ and $j = \langle 0, 1 \rangle$



Cont...

$$\begin{aligned}\text{if } a &= \langle a_1, a_2, a_3 \rangle = \langle a_1, 0, 0 \rangle + \langle 0, a_2, 0 \rangle + \langle 0, 0, a_3 \rangle \\ &= a_1 \langle 1, 0, 0 \rangle + a_2 \langle 0, 1, 0 \rangle + a_3 \langle 0, 0, 1 \rangle = a_1 i + a_2 j + a_3 k\end{aligned}$$

Thus any vector in V_3 can be expressed in terms of i, j, k and for instance,

$$\langle 1, -2, 6 \rangle = i - 2j + 6k$$

Similarly, in two dimensions, we can write

$$a = \langle a_1, a_2 \rangle = a_1 i + a_2 j$$



Example 15

Question

If $a = i + 2j - 3k$ and $b = 4i + 7k$, express the vector $2a + 3b$ in terms of i, j, k .

Solution

Using Properties 1,2,5,6, and 7 of vectors, we have

$$\begin{aligned} 2a + 3b &= 2(i + 2j - 3k) + 3(4i + 7k) \\ &= 2i + 4j - 6k + 12i + 21k = 14i + 4j + 15k \end{aligned}$$

Unit Vector

A **unit vector** is a vector whose **length** is **1**. For instance i, j, k and are all unit vectors. In general, if $a \neq 0$, then the unit vector that has the same direction as a is

$$u = \frac{1}{|a|} a = \frac{a}{|a|} \quad (69)$$



Cont...

In order to verify this , we let $c = \frac{1}{|a|}$. Then $u = ca$ and c is a positive scalar, so u has the same direction as a . Also

$$|u| = |ca| = |c||a| = \frac{1}{|a|}|a| = 1 \quad (70)$$



Example 16

Question

Find the unit vector in the direction of the vector $2i - j - 2k$.

Solution

The given vector has length

$$|2i - j - 2k| = \sqrt{2^2 + (-1)^2 + (-2)^2} = \sqrt{9} = 3$$

So, the unit vector with the same direction is

$$\frac{1}{3}(2i - j - 2k) = \frac{2}{3}i - \frac{1}{3}j - \frac{2}{3}k$$

Dot Product

DEFINITION

If $a = \langle a_1, a_2, a_3 \rangle$ and $b = \langle b_1, b_2, b_3 \rangle$, then the dot product of a and b is the number $a \cdot b$ given by

$$a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3$$

It is a real number, that is, a scalar. For this reason, the **dot product** is sometimes called the **scalar product (or inner product)**.



Cont...

The dot product of two-dimensional vectors is defined in a similar fashion.

$$\langle a_1, a_2 \rangle \cdot \langle b_1, b_2 \rangle = a_1 b_1 + a_2 b_2$$



Example 17

$$\textcircled{1} \quad \langle 2, 4 \rangle \cdot \langle 3, -1 \rangle = 2(3) + 4(-1) = 2$$

$$\textcircled{2} \quad \langle -1, 7, 4 \rangle \cdot \langle 6, 2, -\frac{1}{2} \rangle = (-1)(6) + 7(2) + 4(-\frac{1}{2}) = 6$$

$$\textcircled{3} \quad (i + 2j - 3k) \cdot (2j - k) = 1(0) + 2(2) + (-3)(-1) = 7$$



Properties of The Dot Product

The dot product obeys many of the laws that hold for ordinary products of real numbers. These are stated in the following theorem.



Cont...

If a, b and c are vectors in V_3 and c is a scalar , then

- $a \cdot a = |a|^2$
- $a \cdot b = b \cdot a$
- $a \cdot (b + c) = a \cdot b + a \cdot c$
- $0 \cdot a = 0$
- $(ca) \cdot b = c(a \cdot b) = a \cdot (cb)$
- $a \cdot b = 0$, a and b are mutually perpendicular or $a = 0$ or $b = 0$.



Exercise 4

Proof the above properties as exercise.



Theorem 1

If θ is the angle between the vectors a and b , then

$$a \cdot b = |a||b|\cos\theta$$



Example 18

Question

If the vector a and b have length 4 and 6, and the angle between them is $\pi/3$, find $a \cdot b$

Solution

Using Theorem 1, we have

$$a \cdot b = |a||b| \cos(\pi/3) = 4 \cdot 6 \cdot \frac{1}{2} = 12$$



Cont...

Solution

To find the angle between two vectors we have,

$$\cos \theta = \frac{a \cdot b}{|a||b|} = \frac{2}{3\sqrt{38}}$$

so the angle between a and b is

$$\theta = \cos^{-1} \left(\frac{2}{3\sqrt{38}} \right) = 84^{\circ}$$

Cont...

Two non-zero vectors a and b are called **perpendicular or orthogonal** if the angle between them is $\theta = \pi/2$

$$a \cdot b = |a||b| \cos(\pi/2) = 0$$

and conversely if

$$a \cdot b = 0$$

the $\cos \theta = 0$, so $\theta = \pi/2$. The zero vector is considered to be perpendicular to all vectors.



Theorem 2

Two vectors a and b are orthogonal if and only if $a \cdot b = 0$



Example 19

Question

Show that $2i + 2j - k$ is perpendicular to $5i - 4j + 2k$

Solution

Since

$$(2i + 2j - k) \cdot (5i - 4j + 2k) = 2(5) + 2(-4) + (-1)(2) = 0$$

these vectors are perpendicular.



Projection

The **scalar projection** of b onto a (also called the **component of b along a**) is defined to be the **signed magnitude** of the vector projection, which is the number $|b|\cos\theta$, where θ is the angle between a and b . This is denoted by $\text{comp}_a b$. It's negative if $\pi/2 < \theta \leq \pi$.



Cont...

The equation

$$a \cdot b = |a||b| \cos \theta = |a|(|b| \cos \theta)$$

shows that the dot product of a and b can be interpreted as the length of a times the scalar projection of b onto a . Also,

$$|b| \cos \theta = \frac{a \cdot b}{|a|} = \frac{a}{|a|} \cdot b$$

the component of b along a can be computed by taking the dot product of b with the unit vector in the direction of a .



Cont...

We summarize these ideas as follows.

- 1 Scalar projection of b onto a :

$$\text{comp}_a b = \frac{a \cdot b}{|a|}$$

- 2 Vector projection of b onto a :

$$\text{proj}_a b = \left(\frac{a \cdot b}{|a|} \right) \frac{a}{|a|} = \frac{a \cdot b}{|a|^2} a$$

The vector projection is the scalar projection times the unit vector in the direction of a .



Example 20

Question

Find the scalar projection and vector projection of $b = \langle 1, 1, 2 \rangle$ onto $a = \langle -2, 3, 1 \rangle$.

Solution

Since $|a| = \sqrt{(-2)^2 + 3^2 + 1^2} = \sqrt{14}$, the scalar projection of b onto a is

$$\text{comp}_a b = \frac{a \cdot b}{|a|} = \frac{(-2)(1) + 3(1) + 1(2)}{\sqrt{14}} = \frac{3}{\sqrt{14}}$$

Cont...

Solution

The vector projection is this scalar projection times the unit vector in the direction of a :

$$\text{proj}_a b = \frac{3}{\sqrt{14} |a|} a = \frac{3}{14} a = \left\langle -\frac{3}{7}, \frac{9}{14}, \frac{3}{14} \right\rangle$$



The Cross Product

Given two nonzero vectors $a = \langle a_1, a_2, a_3 \rangle$ and $b = \langle b_1, b_2, b_3 \rangle$, it is very useful to be able to find a nonzero vector that is **perpendicular** to both a and b . If $c = \langle c_1, c_2, c_3 \rangle$ is such a vector, then $a \cdot c = 0$ and $b \cdot c = 0$ and so

$$a_1 c_1 + a_2 c_2 + a_3 c_3 = 0 \quad (71)$$



Cont...

$$b_1 c_1 + b_2 c_2 + b_3 c_3 = 0 \quad (72)$$

To eliminate c_3 we multiply (78) by b_3 and (79) by a_3 and subtract:

$$(a_1 b_3 - a_3 b_1) c_1 + (a_2 b_3 - a_3 b_2) c_2 = 0 \quad (73)$$



Cont...

Equation (80) has the form $pc_1 + qc_2 = 0$, for which an obvious solution is

$c_1 = q$ and $c_2 = -p$.

So a solution of (80) is

$$c_1 = a_2b_3 - a_3b_2, c_2 = a_3b_1 - a_1b_3 \quad (74)$$

Substituting these values into (78) and (79), we then get

$$c_3 = a_1b_2 - a_2b_1$$



Cont...

This means that a vector perpendicular to both a and b is

$$\langle c_1, c_2, c_3 \rangle = \langle a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1 \rangle$$

The resulting vector is called the **cross product** of a and b and it's denoted by $a \times b$



Definition

If $a = \langle a_1, a_2, a_3 \rangle$ and $b = \langle b_1, b_2, b_3 \rangle$, then the cross product of a and b is the vector

$$a \times b = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle \quad (75)$$



Cont...

The cross product $a \times b$ of two vectors a and b , unlike the dot product $a \cdot b$, is a **vector**. For this reason it is also called the **vector product**.

Note that $a \times b$ is defined only when a and b are three-dimensional vectors.



Cont...

In order to make the Definition easier to remember, we use the notation of determinants.

A determinant of order 2 is defined by

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$



Cont...

A determinant of order 3 can be defined in terms of second-order determinants as follows:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix} \quad (76)$$

If we now rewrite the Definition above using second-order determinants and the standard basis vectors i, j and k



Cont...

we see that the cross product of the vectors
 $a = a_1i + a_2j + a_3k$ and $b = b_1i + b_2j + b_3k$ is

$$a \times b = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} i - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} j + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} k \quad (77)$$



Cont...

In view of the similarity between Equation (83) and (84), we often write

$$a \times b = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \quad (78)$$



Example 21

Question

If $a = \langle 1, 3, 4 \rangle$ and $b = \langle 2, -7, -5 \rangle$, then

Solution

$$\begin{aligned} a \times b &= \begin{vmatrix} i & j & k \\ 1 & 3 & 4 \\ 2 & 7 & -5 \end{vmatrix} = \begin{vmatrix} 3 & 4 \\ 7 & -5 \end{vmatrix} i - \begin{vmatrix} 1 & 4 \\ 2 & -5 \end{vmatrix} j + \begin{vmatrix} 1 & 3 \\ 2 & 7 \end{vmatrix} k \\ &= (-15 - 28)i - (-5 - 8)j + (7 - 6)k = -43i + 13j + k \end{aligned}$$



Theorem 3

The vector $a \times b$ is orthogonal to a and b .



Proof

In order to show that $a \times b$ is orthogonal to a , we compute their dot product as follows:

$$\begin{aligned}(a \times b) \cdot a &= a \times b = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} a_1 - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} a_2 + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} a_3 \\&= a_1(a_2b_3 - a_3b_2) - a_2(a_1b_3 - a_3b_1) + a_3(a_1b_2 - a_2b_1) \\&= a_1a_2b_3 - a_1b_2a_3 - a_1a_2b_3 + b_1a_2a_3 + a_1b_2a_3 - b_1a_2a_3 = 0\end{aligned}$$



Cont...

A similar computation shows that $(a \times b) \cdot b = 0$. Therefore $a \times b$ is orthogonal to both a and b .



Theorem 4

If θ is the angle between a and b (so $0 \leq \theta \leq \pi$), then

$$|a \times b| = |a||b| \sin \theta$$

COROLLARY

Two nonzero vectors a and b are parallel if and only if

$$a \times b = 0$$



Proof

Two nonzero vectors a and b are parallel if and only if $\theta = 0$ or π . For $\theta = 0$, $\sin \theta = 0$, and for $\theta = \pi$, $\sin \theta = 0$. Hence, if a and b are parallel, then $\sin \theta = 0$, making $|a \times b| = |a||b|\sin \theta = 0$. Conversely, if $|a \times b| = 0$, then either $|a| = 0$, $|b| = 0$, or $\sin \theta = 0$. However, since $|a|$ and $|b|$ are magnitudes (which are always positive), for $|a \times b|$ to be zero, $\sin \theta$ must be zero, implying $\theta = 0$ or π , and therefore, a and b are parallel.



LEMMA

- 1 The area of a parallelogram with adjacent sides a and b is given by $|a \times b|$
- 2 The area of a triangle with adjacent sides a and b is given by $\frac{1}{2}|a \times b|$
- 3 Two vectors a and b are collinear (linear dependent) if $a \times b = 0$



Example 22

(a) Find a vector perpendicular to the plane that passes through the points $P(1, 4, 6)$, $Q(-2, 5, -1)$, $R(1, -1, 1)$

Ans: $-40i - 15j + 15k$

(a) Find the area of the triangle with vertices $P(1, 4, 6)$, $Q(-2, 5, -1)$, $R(1, -1, 1)$.

Ans: $\frac{5}{2}\sqrt{82}$



Theorem

If a, b, c are vectors and k is a scalar, then

- ① $a \times b = -b \times a$
- ② $(ka) \times b = k(a \times b) = a \times (kb)$
- ③ $a \times (b + c) = a \times b + a \times c$
- ④ $(a + b) \times c = a \times c + b \times c$
- ⑤ $a \cdot (b \times c) = (a \times b) \cdot c$
- ⑥ $a \times (b \times c) = (a \cdot c)b - (a \cdot b)c$



Triple Product

Two products $a.(b \times c)$ is called the **scalar triple product** of the vectors a, b and c . We can write the scalar triple product as a determinant:

$$a.(b \times c) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \quad (79)$$



Application Of Scalar Multiplication

- 1 The volume of a parallel piped with adjacent edges a, b, c is given by $|a.(b \times c)|$.
- 2 The volume of a tetrahedron with adjacent edges a, b, c is given by $\frac{1}{6} |a.(b \times c)|$



Example 23

Use the scalar triple product to show that the vectors $a = \langle 1, 4, -7 \rangle$, $b = \langle 2, -1, 4 \rangle$, and $c = \langle 0, -9, 18 \rangle$ are coplanar.

Ans:0



Equation Of Lines And Planes

A line in the xy - plane is determined when a point on the line and the direction of the line(its slope or angle of inclination) are given.

Likewise, a line L in three-dimensional space is determined when we know a point $P_0(x_0, y_0, z_0)$ on L and the direction of L .



Cont...

Let v be a vector parallel to L .

Let $P(x, y, z)$ be an arbitrary point on L and let r_0 and r be the position vectors of P_0 and P (that is, they have representations $\overrightarrow{OP_0}$ and $\overrightarrow{OP}$).

If a is the vector with representation $\overrightarrow{P_0P}$, then the triangle Law for vector addition gives $r = r_0 + tv$.



Cont...

But , since a and v are parallel vectors, there is a scalar t such that $a = tv$. Thus

$$r = r_0 + tv \quad (80)$$

which is a **vector equation** of L



Equation of Lines

If the vector v that gives the direction of the line L is written in component form as $v = \langle a, b, c \rangle$, then we have $tv = \langle ta, tb, tc \rangle$. We can also write $r = \langle x, y, z \rangle$ and $r_0 = \langle x_0, y_0, z_0 \rangle$, so the vector equation (1) becomes

$$\langle x, y, z \rangle = \langle x_0 + ta, y_0 + tb, z_0 + ct \rangle \quad (81)$$



Cont...

Two vectors are equal if and only if corresponding components are equal.

Therefore we have the three scalar equation:

$$x = x_0 + ta, \quad y = y_0 + tb \quad z =, z_0 + ct \quad (82)$$

where $t \in \text{Re}$. These equations are called **parametric equations** of the line L through the point $P_0(x_0, y_0, z_0)$ and parallel to the vector $v = \langle a, b, c \rangle$. Each value of the parameter t gives a point (x, y, z) on L



Examples 24

Question

- (a) Find a vector equation and parametric equations for the line that passes through the point $(5, 1, 3)$ and is parallel to the vector $i + 4j - 2k$
- (b) Find two other points on the line.



Cont...

Solution

(a) Here $r_0 = \langle 5, 1, 3 \rangle = 5i + j + 3k$ and $v = i + 4j - 2k$, so the vector equation becomes

$$\begin{aligned} r &= (5i + j + 3k) + t(i + 4j - 2k) \\ r &= (5 + t)i + (1 + 4t)j + (3 - 2t)k \end{aligned}$$

Parametric equations are

$$x = 5 + t, y = 1 + 4t, z = 3 - 2t$$



Cont...

Solution

(b) Choosing the parameter value $t = 1$ gives $x = 6, y = 5$ and $z = 1$, so $(6, 5, 1)$ is a point on the line. Similarly, $t = -1$ gives the point $(4, -3, 5)$.



Equation of lines

The vector equation and parametric equations of a line are not unique.

In general , if a vector $v = \langle a, b, c \rangle$ is used to describe the direction of a line L , then the numbers a, b, c are called **direction numbers of L** .



Cont...

If none of a, b, c is 0, we solve each of these equations for t , equate the results, and obtain

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c} \quad (83)$$

These equations are called **symmetric equations** of L .

If one of a, b, c is zero, we can still eliminate t . For instance, if $a = 0$, we could write the equations of L as

$$x - x_0 = \frac{y - y_0}{b} = \frac{z - z_0}{c} \quad (84)$$

This means that L lies in the vertical plane $x = x_0$



Example 25

Question

- (a) Find parametric equations and symmetric equations of the line that passes through the points $A(2, 4, -3)$ and $B(3, -1, 1)$.
- (b) At what point does this line intersect the xy - plane?



Cont...

Solution

(a) We are not explicitly given a vector parallel to the line, but observe that the vector v with representation $\overrightarrow{AB}$ is parallel to the line and

$$v = \langle 3 - 2, -1 - 4, 1 - (-3) \rangle = \langle 1, -5, 4 \rangle$$

Thus direction numbers are $a = 1, b = -5$, and $c = 4$.



Cont...

Solution

Taking the point $(2, 4, -3)$ as P_0 we see that parametric equation are

$$x = 2 + t, y = 4 - 5t, z = -3 + 4t \quad (85)$$



Cont...

Solution

and symmetric equations are

$$\frac{x-2}{1} = \frac{y-4}{-5} = \frac{z+3}{4}$$

or by the corresponding vector equation

$$r(t) = \langle 2+t, 4-5t, -3+4t \rangle, 0 \leq t \leq 1 \quad (86)$$



Cont...

Solution

(b) The line intersects the xy - plane when $z = 0$ in the symmetric equations and obtain

$$\frac{x-2}{1} = \frac{y-4}{-5} = \frac{3}{4} \quad (87)$$

This gives $x = \frac{11}{4}$ and $y = \frac{1}{4}$, so the line intersects the xy - plane at the point $(\frac{11}{4}, \frac{1}{4}, 0)$.



Equations of Lines

In general, we know that the vector equation of a line through the (tip of the) vector r_0 in the direction of a vector v is $r = r_0 + tv$. If the line also passes through (the tip of) r_1 , then we can take $r_1 - r_0$ and so its vector equation is

$$r = r_0 + t(r_1 - r_0) + tr_1 \quad (88)$$



Cont...

The line segment from r_0 to r_1 is given by the parameter interval

$$r = r_0 + t(r_1 - r_0), 0 \leq t \leq 1 \quad (89)$$



Example 26

Question

Show that the lines L_1 and L_2 with parametric equations

① $x = 1 + t, y = -2 + 3t, z = 4 - t$

② $x = 2s, y = 3 + 4s, z = -3 + 4s$

are skew lines, that is, they do not intersect and are not parallel (and therefore do not lie in the same plane)



Cont...

Solution

The lines are not parallel because the corresponding vectors $\langle 1, 3, -1 \rangle$ and $\langle 2, 1, 4 \rangle$ are not parallel. (Their components are not proportional). If L_1 and L_2 has a point of intersection, there would be values of t and s such that

$$1 + t = 2s$$

$$-2 + 3t = 3 + s$$

$$4 - t = -3 + 4s$$

Cont...

But if we solve the first two equations, we get $t = \frac{11}{5}$ and $s = \frac{8}{5}$, and these values don't satisfy the third equation. Therefore, there are no values of t and s that satisfy the three equations, so L_1 and L_2 don't intersect. Thus, L_1 and L_2 are skew lines.



Equations of Planes

Let $P(x, y, z)$ be an arbitrary point in the plane and let r_0 and r be the position vectors of P_0 and P . Then the vector $r - r_0$ is represented by $\overrightarrow{P_0P}$. The normal vector n is orthogonal to every vector in the given plane.

In particular, n is orthogonal to $r - r_0$ and so we have

$$n \cdot (r - r_0) = 0 \quad (90)$$



Cont...

which can be rewritten as

$$n \cdot r = n \cdot r_0 \quad (91)$$

Either Equation (90) or (91) is called a vector equation of the plane. To obtain a scalar equation for the plane, we write $n = \langle a, b, c \rangle$, $r = \langle x, y, z \rangle$ and $r_0 = \langle x_0, y_0, z_0 \rangle$.



Equation of Planes

Then the vector equation (90) becomes

$$\langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0 \quad (92)$$

or

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0 \quad (93)$$

Equation (93) is the scalar equation of the plane through $P_0(x_0, y_0, z_0)$ with normal vector $n = \langle a, b, c \rangle$



Example 27

Question

Find an equation of the plane through the point $(2, 4, -1)$ with normal vector $n = \langle 2, 3, 4 \rangle$. Find the intercepts.



Cont...

Solution

Putting $a = 2, b = 3, c = 4, x_0 = 2, y_0 = 4$, and $z_0 = -1$ in Equation (100), we see that an equation of the plane is

$$2(x - 2) + 3(y - 4) + 4(z + 1) = 0$$

which simplifies to

$$2x + 3y + 4z = 12$$



Cont...

Solution

To find the x -intercept we set $y = z = 0$ in this equation and obtain $x = 6$. Similarly, the y -intercept is 4 and the z -intercept is 3. By collecting terms in Equation (100) as we did, we can rewrite the equation of the plane as

$$ax + by + cz + d = 0$$

where $d = -(ax_0 + by_0 + cz_0)$. This equation is called a **linear equation** in x, y, z .

Example 28

Question

Find an equation of the plane that passes through the points $P(1,3,2)$, $Q(3,-1,6)$ and $R(5,2,0)$

Solution

The vectors a and b corresponding to $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ are

$$a = \langle 2, -4, 4 \rangle, b = \langle 4, -1, -2 \rangle$$

Cont...

Solution

Since both a and b lie in the plane, their cross product $a \times b$ is orthogonal to the plane and can be taken as the normal vector. Thus

$$n = a \times b = \begin{vmatrix} i & j & k \\ 2 & -4 & 4 \\ 4 & -1 & -2 \end{vmatrix} = 12i + 20j + 14k \quad (94)$$



Cont...

Solution

With the point $P(1,3,2)$ and the normal vector $n = 12i + 20j + 14k$, an equation of the plane is

$$12(x - 1) + 20(y - 3) + 14(z - 2) = 0 \quad (95)$$

or

$$6x + 10y + 7z = 50 \quad (96)$$



Exercise 5

- (a) Find the angle between the plane $x + y + z = 1$ and $x - 2y + 3z = 1$ and Find the symmetric equations for the line of intersection L of these two planes.
- (b) Find the distance between the parallel planes $10x + 2y - 2z = 5$ and $5x + y - z = 1$



Example 29

Question

Find a formula for the distance D from a point $P_1(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$

Solution

Let $P_0(x_0, y_0, z_0)$ be any point in the given plane and let b be the vector corresponding to $\overrightarrow{P_0P_1}$. Then

$$b = \langle x_1 - x_0, y_1 - y_0, z_1 - z_0 \rangle \quad (97)$$

Cont...

Solution

We can see that the distance D from P_1 to the plane is equal to the absolute value projection of b onto the normal vector $n = \langle a, b, c \rangle$

$$D = |comp_n b| = \frac{|n \cdot b|}{|n|} \quad (98)$$



Cont...

Solution

$$= \frac{|a(x_1 - x_0) + b(y_1 - y_0) + c(z_1 - z_0)|}{\sqrt{a^2 + b^2 + C^2}} \quad (99)$$

$$= \frac{|(ax_1 + by_1 + cz_1) - (ax_0 + by_0 + cz_0)|}{\sqrt{a^2 + b^2 + C^2}} \quad (100)$$



Cont...

Solution

Since P_0 lies in the plane, its coordinates satisfy the equation of the plane and so we have $ax_0 + by_0 + cz_0 + d = 0$. Thus the formula for D can be written as

$$D = \frac{|(ax_1 + by_1 + cz_1 + d)|}{\sqrt{a^2 + b^2 + C^2}} \quad (101)$$



Vector Space

Let V be a set on which addition and scalar multiplication are defined (this means that if u and v are objects in V and c is a scalar , then we've defined $u + v$ and cu in some way) . If the following axioms are true for all objects u, v , and w in V and all scalars c and k , then V is called a **vector space** and the **objects in V** are called **vectors**.



Properties Of Vector Space

- (a) $u + v$ is in V - This is called closed under addition
- (b) cu is in V - This is called closed under multiplication.
- (c) $u + v = v + u$
- (d) $u + (v + w) = (u + v) + w$
- (e) There is a special object in V , denoted 0 and called the zero vector, such that for all u in V we have $u + 0 = 0 + u = u$.



Cont...

(f) For every u in V there is another object in V , denoted $-u$ and called the negative of u , such that $u - u = u + (-u) = 0$.

(g) $c(u + v) = cu + cv$

(h) $(c + k)u = cu + ku$

(i) $c(ku) = (ck)u$

(j) $1u = u$



Linear Combination

We say that the vector w from the vector space V is a linear combination of the vectors $v_1, v_2, \dots, v_n$ all from V if there are scalars $c_1, c_2, \dots, c_n$ so that w can be written

$$w = c_1 v_1 + c_2 v_2 + \dots + c_n v_n \quad (102)$$



Example 30

Question

Determine if the vector is a linear combination of the two given vectors.

1. Is $w = (-12, 20)$ a linear combination of $v_1 = (-1, 2)$ and $v_2 = (4, -6)$?
2. Is $w = (1, -4)$ a linear combination of $v_1 = (2, 10)$ and $v_2 = (-3, -15)$?



Cont...

Solution

In each of these cases we'll need to set up and solve the following equation,

$$w = c_1 v_1 + c_2 v_2 \quad (103)$$



Cont...

Solution

1.

$$(-12, 20) = c_1(-1, 2) + c_2(4, -6)$$

Equating the coefficients gives

$$-c_1 + 4c_2 = -12$$

$$2c_1 - 6c_2 = 20$$



Cont...

Solution

If the system is consistent (i.e has at least one solution) then w is a linear combination of the two vectors, else w is not a linear combination of the two vectors.

Therefore w is a linear combination of v_1 and v_2 and we can write $w = 4v_1 - 2v_2$ when $c_1 = 4$ and $c_2 = -2$



Cont...

Solution

2. Here the system will need is

$$2c_1 - 3c_2 = 1$$

$$10c_1 - 15c_2 = -4$$

This system does not have a solution and so w is not a linear combination.



Definition

Suppose $S = v_1, v_2, \dots, v_n$ is a non-empty set of vectors and form the vector equation,

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0 \quad (104)$$

This equation has at least one solution, namely

$$c_1 = 0, c_2 = 0, \dots, c_n = 0.$$

This solution is called the **trivial solution**.



Cont...

If the trivial solution is the only solution to this equation then the vectors in the set S are called linearly independent and the set is called a linearly independent set.

If there is another solution then the vectors in the set S are called linearly dependent and the set is called a linearly dependent set.



Exercise 6

Determine if each of the following sets are linearly independent or linearly dependent.

- 1 $v_1 = (3, -1)$ and $v_2 = (-2, 2)$
- 2 $v_1 = (12, -8)$ and $v_2 = (-9, 6)$
- 3 $v_1 = (1, 0, 0)$, $v_2 = (0, 1, 0)$ and $v_3 = (0, 0, 1)$
- 4 $v_1 = (2, -2, 4)$, $v_2 = (3, -5, 4)$ and $v_3 = (0, 1, 1)$



Linear Independence and Dependence

Theorem 1: Simplest linear independent set

A single nonzero vector v is linearly independent.

Proof: Suppose $kv = 0$, then $k = 0$. Since $v \neq 0$, according to the definition, v must be linearly independent.

Theorem 2: Simplest linear dependent set

The zero vector 0 is linearly dependent.

Proof: $ck = 0$ does not necessarily imply that $k = 0$. According to the definition, 0 must be linearly dependent.

Definition

Vectors $v_1, v_2, \dots, v_n$ all from V are said to constitute a basis for w if

- 1 They are linearly independent
- 2 For every vector $w \in V$, there exist scalars $c_1, c_2, \dots, c_n$ so that w can be written

$$w = c_1 v_1 + c_2 v_2 + \dots + c_n v_n \quad (105)$$

$c_1, c_2, \dots, c_n$ are called coordinates of w relative to the given basis.



Matrix Algebra

A matrix is a rectangular array of numbers arranged in m horizontal rows and n vertical columns. For example,

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \quad (106)$$

We say that A is an $m \times n$ matrix. a_{ij} in the i th row and j th column of A is called the (i, j) th element of A .



Cont...

Two matrices A and B are equal if $a_{ij} = b_{ij}$ for all i, j . If $m = n$, then A is called a square matrix of order n .

$a_{ii}, i = 1, 2, \dots, n$ are called the diagonal elements of the square matrix. A square matrix A such that $a_{ij} = 0$ for $i \neq j$ is called a diagonal matrix.



Addition Of Matrices

Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ be two matrices. The sum is defined as

$$A + B = C$$

where $C = [c_{ij}]_{m \times n}$ and $c_{ij} = a_{ij} + b_{ij}$. That is C is obtained by adding corresponding elements of A and B .



Cont...

For Instance,

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 4 & 0 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 4 & 2 \\ 1 & -1 & 0 \end{bmatrix}$$

Then

$$A + B = \begin{bmatrix} 5 & 7 & 1 \\ 5 & -1 & 3 \end{bmatrix}$$



Cont...

observe that the matrix sum or difference is defined only when the two matrices have the same number of rows and same number of columns.

A matrix such that all its entries are zero is called a zero matrix and is denoted by 0 . That is in a zero matrix $a_{ij} = 0$ for all i and j .



Properties of Matrix Addition

Let A, B, C be three matrix of order $m \times n$ is a null matrix of the same order, then

- ① $A + B = B + A$
- ② $(A + B) + C = A + (B + C)$
- ③ $A + 0 = 0 + A = A$



Multiplication Of Matrices

MULTIPLICATION OF A MATRIX BY A SCALAR

Let $A = [a_{ij}]_{m \times n}$ and α is a scalar (a real number), then $\alpha A = [\alpha a_{ij}]_{m \times n}$ is called the scalar multiplication of A by α .



Cont...

MULTIPLICATION OF MATRIX BY A MATRIX

Let $A = [a_{ij}]$ be an $m \times p$ matrix and $B = [b_{ij}]$ be a $p \times m$ matrix. Then the product of A and B denoted by AB is the $m \times n$ matrix $C = [c_{ij}]$, where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ip}b_{pj}$$



Cont...

MULTIPLICATION OF MATRIX BY A MATRIX

for $1 \leq i \leq m, 1 \leq j \leq n$. c_{ij} can be computed in the following ways:

- 1 Select i th row in A and j th column in B and place them side by side.
- 2 Multiply the corresponding entries and add all the products.



Cont...

C is defined only when the number of columns in A is equal to the number of rows in B . We write $A.B$ only when it is defined. If $A.B$ is defined then $B.A$ may not be defined.



Example 31

Question

If A and B are two matrices of order 2×3 and 3×2 respectively, such that

$$A = \begin{bmatrix} 2 & 1 & 23 \\ 4 & 1 & -2 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 3 \end{bmatrix}$$



Cont...

Solution

Then

$$AB = \begin{bmatrix} 7 & 13 \\ -3 & 0 \end{bmatrix}$$

$$BA = \begin{bmatrix} 4 & 1 & -2 \\ 10 & 3 & -1 \\ 16 & 5 & 0 \end{bmatrix}$$



Properties Of Matrix Multiplication

Let A , B , and C be three matrices of order $m \times n$. Then

- ① $A \cdot (B \cdot C) = (A \cdot B) \cdot C$
- ② $A \cdot (B + C) = A \cdot B + A \cdot C$
- ③ $(A + B) \cdot C = A \cdot C + B \cdot C$
- ④ The $n \times n$ diagonal matrix, where $a_{ij} = 1$ for all i on the diagonal, is called the identity matrix of order n and is denoted by I_n



Cont...

For example, $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- ④ If A is an $n \times n$ matrix, then $A \cdot I_n = I_n \cdot A = A$
- ⑤ If A is a square matrix of order n , then we define $A^p = A \cdot A \cdot \dots \cdot A$ (p factors) and $A^0 = I$, then we have $A^p \cdot A^q = A^{p+q}$ for non-negative integers p and q .



Transpose of a Matrix

If $A = [a_{ij}]$ is an $m \times n$ matrix, then the $n \times m$ matrix $A^T = [a_{ji}]$, where $a_{ji} = a_{ij}$ for $1 \leq i \leq m$, $1 \leq j \leq n$, is called the transpose of A . For example, if

$$A = \begin{bmatrix} 1 & -3 & 4 \\ 5 & 6 & -2 \end{bmatrix} \qquad A^T = \begin{bmatrix} 1 & 5 \\ -3 & 6 \\ 4 & -2 \end{bmatrix}$$

That is, A^T is obtained by interchanging rows and columns of A .



Properties of Transpose

If A and B are two matrices then

$$① \quad (A^T)^T = A$$

$$② \quad (A + B)^T = A^T + B^T$$

$$③ \quad (AB)^T = B^T A^T$$



Trace

If A is a square matrix, then the trace of A denoted by $\text{tr}(A)$ is defined by the **entries of the main diagonal of A** . The trace of A is undefined if A is not a square matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{tr}(A) = a_{11} + a_{22} + a_{33}$$



Example 32

If

$$A = \begin{bmatrix} 4 & 1 & -2 \\ 10 & 3 & -1 \\ 16 & 5 & 0 \end{bmatrix}$$

then $tr(A) = 4 + 3 + 0 = 7$



Symmetric Matrix

A matrix $A = [a_{ij}]$ is called **symmetric** if $A = A^T$. That is A is symmetric if and only if $a_{ij} = a_{ji}$, for $1 \leq i \leq m, 1 \leq j \leq n$. If $A = -A^T$ then A is called **skew symmetric**.



Example 33

For,

$$A = \begin{bmatrix} 2 & -3 & 4 \\ -3 & 1 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

These matrices are symmetric.

If A is symmetric, then for all positive integers n , A^n is symmetric.



Cont...

For,

$$A = \begin{bmatrix} 0 & 2 & 1 \\ -2 & 0 & -3 \\ -1 & 3 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$$

These matrices are skew symmetric.



Theorem

Let A and B be symmetric matrices of the same size and C is any scalar , then

- 1 $A + B$ is symmetric
- 2 $A - B$ is symmetric.
- 3 AB is symmetric if and only if $AB = BA$
- 4 cA is symmetric.
- 5 A^T is symmetric.



Theorem

Let A and B be skew symmetric matrices and scalar k , then

- 1 $A + B$ is skew symmetric.
- 2 $A - B$ is skew symmetric
- 3 kA is skew symmetric.
- 4 If A is $n \times n$ matrix, then $A - A^T$ is skew symmetric.
- 5 $A^T = -A$, then BAB^T is skew symmetric matrix for all B .



Unitary Matrix

A square matrix A with the property that $A^{-1}A$ is to be unitary.

EXAMPLE

$$A = \begin{bmatrix} i & 0 \\ 0 & 1 \end{bmatrix}$$



Skew Hermitian Matrix

A matrix is said to be **skew hermitian** if $\overline{A}^T = -A$

EXAMPLE

$$A = \begin{bmatrix} i & 2+i & 3+2i \\ -2+i & 3i & -3i \\ -3+2i & -3i & 0 \end{bmatrix}$$



Orthogonal Matrix

A square matrix with the property that $A^{-1} = A^T$ is said to be orthogonal. It follows from this definition that a square matrix A is **orthogonal** if and only if $AA^T = A^T A = I$

EXAMPLE

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$$



Hermitian Matrices

A matrix is said to be **hermitian** if $A = \overline{A}^T$

EXAMPLE

$$A = \begin{bmatrix} 2 & 1-i & 0 \\ 1+i & -1 & i \\ 0 & -i & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 2+i \\ 2-i & -12 \end{bmatrix}$$



Inverse Matrices

If A is a square matrix and we can find another matrix of the same size, say B , such that

$$AB = BA = I$$

then we call A **invertible** and we say that B is an **inverse of the matrix A** . If we can't find such a matrix B , we call A a **singular matrix** ($\det(A) = 0$).



Theorem

Suppose that A is invertible and that both B and C are inverses of A , then $B = C$ and we will denote the inverse as A^{-1} .



Example 34

Question

Given the matrix A , verify that the indicated matrix is in fact the inverse.

$$A = \begin{bmatrix} -4 & -2 \\ 5 & 5 \end{bmatrix}, \quad A^{-1} = \begin{bmatrix} -1/4 & -1/5 \\ 1/2 & 2/5 \end{bmatrix}$$



Cont...

Solution

To verify that we do, in fact, have the inverse, we'll need to check that $AA^{-1} = A^{-1}A = I$. This is easy enough to do, and so we'll leave it to you to verify the multiplication.

$$AA^{-1} = \begin{bmatrix} -4 & -2 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} -1/4 & -1/5 \\ 1/2 & 2/5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



Definition

If A is a square matrix and $n > 0$ then,

$$A^{-n} = (A^{-1})^n = A^{-1} A^{-1} \dots A^{-1}$$



Theorem

Suppose that A and B are invertible matrices of the same size. Then,

- 1 AB is invertible and $(AB)^{-1} = B^{-1}A$
- 2 A^{-1} is invertible and $(A^{-1})^{-1} = A$
- 3 For $n = 0, 1, 2, \dots$, A^n is invertible and $(A^n)^{-1} = A^{-n} = (A^{-1})^n$.
- 4 If c is a non - zero scalar cA is invertible and $(cA)^{-1} = \frac{1}{c}A^{-1}$.
- 5 A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$



Finding Inverse Matrices

If A is an $n \times n$ matrix then the following statements are equivalent

- 1 A is invertible
- 2 The only solution to the system $Ax = 0$ is the trivial solution.
- 3 A is row equivalent to I_n .
- 4 $Ax = b$ has exactly one solution for every $n \times 1$ matrix b .
- 5 $Ax = b$ is consistent for every $n \times 1$ matrix b



Theorem

The matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible if $ad - bc \neq 0$ and singular if $ad - bc = 0$.

If the matrix is invertible its inverse will be,

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$



Example 35

Question

Use the fact to show that

$$A = \begin{bmatrix} -4 & -2 \\ 5 & 5 \end{bmatrix}$$

is an invertible matrix and find its inverse.



Cont...

Solution

First, we need,

$$ad - bc = (-4)(5) - (5)(-2) = -10 \neq 0$$

The matrix is, in fact, invertible and here is the inverse,

$$A^{-1} = \frac{1}{-10} \begin{bmatrix} 5 & 2 \\ -5 & -4 \end{bmatrix} = \begin{bmatrix} -1/2 & -1/5 \\ 1/2 & 2/5 \end{bmatrix}$$



Example 36

Question

Determine if the following matrix is singular.

$$A = \begin{bmatrix} -4 & -2 \\ 6 & 3 \end{bmatrix}$$

Solution

$$(-4)(3) - (-2)(6) = 0$$

the matrix is singular.

Determinant

If A is square matrix then the determinant function is denoted by $\det$ and $\det(A)$ is defined to be the sum of all the signed elementary products of A .

Note that often we will call the number $\det(A)$ the determinant of A . Also, there is some alternate notation that is sometimes used for determinants. We will sometimes denote determinants as $\det(A) = |A|$ and this is most often done with the actual matrix instead of the letter representing the matrix.



Determinant function for a 2×2 matrix

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$



cont...

Determinant function for a 3×3 matrix

$$\begin{aligned}\det(A) &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\ &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}\end{aligned}$$



Example 37

Question

Compute the determinant of each of the following matrices.

$$A = \begin{bmatrix} 3 & 2 \\ -9 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 5 & 4 \\ -2 & -1 & 8 \\ -11 & 1 & 7 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & -6 & 2 \\ 2 & -8 & 3 \\ -3 & 1 & 1 \end{bmatrix}$$



Cont...

Solution

$$\det(A) = (3)(5) - (2)(-9) = 33$$

$$\det(B) = 3 \begin{vmatrix} -1 & 8 \\ 1 & 7 \end{vmatrix} - 5 \begin{vmatrix} -2 & 8 \\ -11 & 7 \end{vmatrix} + 4 \begin{vmatrix} -2 & -1 \\ -11 & 1 \end{vmatrix}$$

Solve C as an Exercise.



Properties Of Determinant

THEOREM 1

Let A be an $n \times n$ matrix and c be a scalar, then,

$$\det(cA) = c^n \det(A)$$



Example 38

Question

For the given matrix below, compute both $\det(A)$ and $\det(2A)$.

$$A = \begin{bmatrix} 4 & -2 & 5 \\ -1 & -7 & 10 \\ 0 & 1 & -3 \end{bmatrix}$$



Cont...

Solution

First the scalar multiple is given by

$$2A = \begin{bmatrix} 8 & -4 & 10 \\ -2 & -14 & 20 \\ 0 & 2 & -6 \end{bmatrix}$$

$$\det(A) = 45, \quad \det(2A) = 2^3 \det(A)$$



Theorem 2

Suppose that A, B and C are all $n \times n$ matrices and that they differ by only a row, say the k^{th} row. Let's further suppose that the k^{th} row of C can be found by adding the corresponding entries from the k^{th} rows of A and B . Then in this case we will have that

$$\det(C) = \det(A) + \det(B)$$



Example 39

Question

Consider the following three matrices.

$$A = \begin{bmatrix} 4 & 2 & -1 \\ 6 & 1 & 7 \\ -1 & -3 & 9 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 2 & -1 \\ -2 & -5 & 3 \\ -1 & -3 & 9 \end{bmatrix} \quad C = \begin{bmatrix} 4 & 2 & -1 \\ 4 & -4 & 10 \\ -1 & -3 & 9 \end{bmatrix}$$



Cont...

Solution

First notice that we can write C as,

$$C = \begin{bmatrix} 4 & 2 & -1 \\ 4 & -4 & 10 \\ -1 & -3 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 2 & -1 \\ 6 + (-2) & 1 + (-5) & 7 + 3 \\ -1 & -3 & 9 \end{bmatrix}$$



Cont...

Solution

All three matrices differ only in the second row and the second row of C can be found by adding the corresponding entries from the second row of A and B . The determinants of these matrices are

$$\det(A) = 15, \det(B) = -115 \text{ and } \det(C) = -100 = 15 + (-115)$$



Theorem 3

If A and B are matrices of the same size then

$$\det(AB) = \det(A)\det(B)$$



Example 40

Question

For the given matrices , compute $\det(A)$, $\det(B)$, and $\det(AB)$

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 7 & 4 \\ 3 & 1 & 4 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 & 8 \\ 4 & -1 & 1 \\ 0 & 3 & 3 \end{bmatrix}$$



Cont...

Solution

$$AB = \begin{bmatrix} -8 & 12 & 15 \\ 28 & 7 & 35 \\ 4 & 14 & 37 \end{bmatrix}$$

$$\det(A) = -41, \quad \det(B) = 84$$

$$\det(AB) = -3444 = (-41)(84) = \det(A)\det(B)$$



Theorem 4

A square matrix A is invertible if and only if $\det(A) \neq 0$. A matrix that is invertible is often called non-singular and a matrix that is not invertible is often called singular.



Theorem 5

If A is a square matrix then,

$$\det(A) = \det(A^T)$$



Example 41

Question

Compute $\det(A)$ and $\det(A^T)$ for the following matrix.

$$A = \begin{bmatrix} 5 & 3 & 2 \\ -1 & -8 & -6 \\ 0 & 1 & 1 \end{bmatrix}$$



Cont...

Solution

You will verify that

$$\det(A) = \det(A^T) = -9$$



Theorem 6

If A is a square matrix with a row or column of all zeroes then

$$\det(A) = 0$$

and so A will be singular.



Theorem 7

Suppose that A is an $n \times n$ triangular matrix (and hence any diagonal), then,

$$\det(A) = a_{11}a_{22}\dots a_{nn}$$



Example 42

Question

Compute the determinant of each of the following matrices.

$$A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 6 & 0 \\ 2 & -1 \end{bmatrix}$$



Cont...

Solution

$$\det(A) = -60$$

$$\det(B) = -6$$



The Method Of Cofactors

Definition 1

If A is a square matrix, then the minor of a_{ij} , denoted by M_{ij} , is the determinant of the submatrix that results from removing the i^{th} row and j^{th} column of A .

Definition 2

If A is a square matrix, then the cofactor of a_{ij} , denoted by C_{ij} , is the number $(-1)^{i+j}M_{ij}$. Cofactors can be used to determine the inverse of an invertible matrix.

Cont...

Definition 3

Let A be an $n \times n$ matrix and C_{ij} be the cofactors of a_{ij} . The matrix of cofactors from A is

$$\begin{bmatrix} C_{11} & \cdots & C_{1n} \\ \vdots & \ddots & \vdots \\ C_{m1} & \cdots & C_{mn} \end{bmatrix}$$

The adjoint of A is the transpose of the matrix of cofactors and is denoted by $adj(A)$.

Example 43

Question

Compute the adjoint of the following matrix.

$$A = \begin{bmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{bmatrix}$$



Cont...

We need the cofactors for each of the entries from this matrix. $A_{11} = (-1)^2 \begin{bmatrix} 6 & 3 \\ -4 & 0 \end{bmatrix} = 12$, $A_{12} = (-1)^3 \begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix} =$

$$6, A_{13} = (-1)^4 \begin{bmatrix} 1 & 6 \\ 2 & -4 \end{bmatrix} = -16$$

$$A_{21} = (-1)^3 \begin{bmatrix} 2 & -1 \\ -4 & 0 \end{bmatrix} = 4, A_{22} = -(1)^4 \begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix} = 2, A_{23} =$$

$$(-1)^5 \begin{bmatrix} 3 & 2 \end{bmatrix}$$



Cont

The matrix of the cofactors is

$$C = \begin{bmatrix} 12 & 6 & -16 \\ 4 & 2 & -10 \\ 12 & -10 & 16 \end{bmatrix}$$

The Adjoint of A is the transpose of the cofactors.

$$\text{adj}(A) = \begin{bmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & -10 & 16 \end{bmatrix}$$



Theorem 8

If A is an invertible matrix then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$



Example 44

Question

Use the adjoint matrix to compute the inverse of the following matrix.

$$A = \begin{bmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{bmatrix}$$



Cont...

Solution

$$\det(A) = 64$$

$$\operatorname{adj}(A) = \begin{bmatrix} 12 & 4 & 12 \\ 6 & 2 & 16 \\ -16 & 16 & 16 \end{bmatrix}$$



Cont...

Solution

Therefore , the inverse of the matrix is,

$$A^{-1} = \frac{1}{64} \begin{bmatrix} 12 & 4 & 12 \\ 6 & 2 & 16 \\ -16 & 16 & 16 \end{bmatrix}$$



Theorem 9

Let A be a square matrix.

- (a) If B is the matrix that results from multiplying a row or column of A by a scalar, c , then $\det(B) = c \det(A)$.
- (b) If B is the matrix that results from interchanging two rows or two columns of A then $\det(B) = -\det(A)$.
- (c) If B is the matrix that results from adding a multiple of row of A onto another row of A or adding a multiple of one column of A onto another column of A then $\det(B) = \det(A)$.



Systems Of Equations

The general form of linear equation is

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where there are n unknowns $,x_1, x_2, \dots, x_n$ and $a_1, a_2, \dots, a_n, b$ are all known numbers. Here are a couple of examples of linear equations.



Note

- (a) The unknowns only appear to the first power and there aren't any unknowns in the denominator of the fraction.
- (b) No products and / or quotients of unknowns.
- (c) Unknowns only occur in the numerators, they are only to the first power and there are no products or quotients of unknowns.



Solution set of a single linear equation

A solution set (or often just a solution) is a set of numbers $t_1, t_2, \dots, t_n$ so that if we set $x_1 = t_1, x_2 = t_2, \dots, x_n = t_n$ the n linear equation above is satisfied.



Example 45

Question

Find the solution set for each of the following linear equations.

(a) $7x_1 - \frac{5}{9}x_2 = -1$

(b) $6x + 8y + 10z = 3$



Cont...

Solution

(a) Finding x_1

$$7x_1 = \frac{5}{9}x_2 - 1 \implies x_1 = \frac{5}{63}x_2 - \frac{1}{7}$$

Let $x_2 = t$

$$\therefore x_1 = \frac{5}{63}t - \frac{1}{7}$$

Cont...

Solution

Now choosing any value of t and plug in to get a pair of numbers x_1 and x_2 that will satisfy the equation.

$$t = 0; x_1 = -\frac{1}{7}; x_2 = 0$$

$$t = 27; x_1 = \frac{5}{63}(27); x_2 = 27$$

Cont...

Solution

We can easily check that these are in fact solutions to the equation by plugging them back into the equation

$$t = 0; 7\left(-\frac{1}{7}\right) - \frac{5}{9}(0) = -1$$

$$t = 27; 7(2) - \frac{5}{9}(27) = -1$$

Cont...

Solution

(b) We will do this with a little less detail since it works in essentially the same manner.

$$10z = 3 - 6x + 8y$$

$$z = \frac{3}{10} - \frac{3}{5}x + \frac{4}{5}y$$

Cont...

Solution

In this case we will need to know values for both x and y in order to get a value for z . As with the first case, there is nothing in this problem to restrict out choices of x and y . The solution set to this linear equation is then,

$$x = t; y = s; z = \frac{3}{10} - \frac{3}{5}x + \frac{4}{5}y$$



Cont...

Solution

So if we choose any values for t and s we can get a set of number solutions as follows.

$$x = 0; y = -2; z = \frac{3}{10} - \frac{3}{5}(0) + \frac{4}{5}(-2) = -\frac{13}{10}$$

$$x = -\frac{3}{2}; y = 5; z = \frac{3}{10} - \frac{3}{5}\left(-\frac{3}{2}\right) = \frac{26}{5}$$



Cont...

Solution

As with the first part if we take either set or three numbers , we can plug them into the equation to verify that the equation will be satisfied.

$$6\left(\frac{-3}{2}\right) - 8(5) + 10\left(\frac{26}{5}\right) = -9 - 40 + 52 = 3$$



A System of Linear Equations

A system of linear equations is nothing more than a collection of two or more linear equations. Here are some examples of systems of linear equations.

$$2x + 3y = 9$$

$$x - 2y = -13$$

$$2x_1 + 3x_2 + 3x_3 = 9$$

$$x_1 + 9x_2 + x_3 = 26$$

$$7x_1 - 5x_2 + 2x_3 = 29$$



Theorem 1

Given a system of n equations and m unknowns, there will be one of three possibilities for solutions to the system.

- 1 There will be no solution.
- 2 There will be exactly one solution.
- 3 There will be infinitely many solutions.

If there is no solution to the system, we call the system **inconsistent**. If there is at least one solution to the system, we call it **consistent**.



Cont...

We will need the following general system of n equations and m unknowns.

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2m}x_m = b_2$$

$$\vdots + \vdots + \dots + \vdots = \vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nm}x_m = b_n$$

In this system the unknowns are $x_1, x_2, \dots, x_m$ and a_{ij} and b_i are known numbers.



Cont...

Below is the augmented matrix for the general system

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1m} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2m} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} & b_n \end{array} \right]$$

Each row of the augmented matrix consists of the coefficients and constant on the right of the equal sign from a given equation in the system.



Cont...

The first row is for the first equation, the second row is for the second equation and so on.

Likewise each of the first m columns of the matrix consists of the coefficients from the unknowns. The first column contains the coefficients of x_1 , the second column contains the coefficients of x_2 , etc. The final column contains all the constants on the right of the equal sign.



cont...

The augmented part of the name arises because we tack the b_i 's onto the matrix. If we don't tack those on then we just have

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}$$

And we call this coefficient matrix for the system.



Example 46

Question

Write down the augmented matrix for the following system.

$$3x_1 - 10x_2 + 6x_3 - x_4 = 3$$

$$x_1 + 9x_3 - 5x_4 = -12$$

$$-4x_1 + x_2 - 9x_3 + 2x_4 = 7$$



Cont...

Solution

There really isn't too much to do here other than write down the system as

$$\begin{bmatrix} 3 & -10 & 6 & -1 & 3 \\ 1 & 0 & 9 & -5 & -12 \\ -4 & 1 & -9 & 2 & 7 \end{bmatrix}$$

Notice that the second equation did not contain an x_2 and so we consider its coefficient to be zero.

Example 47

Question

For the given augmented matrix write down the corresponding system of equations.

$$\left[\begin{array}{ccc|c} 4 & -1 & 1 & \\ -5 & -8 & 4 & \\ 9 & 2 & -2 & \end{array} \right]$$



Cont...

Solution

Here is the system that corresponds to this augmented matrix.

$$\begin{array}{rcl} 4x_1 & - & x_2 = 1 \\ -5x_1 & - & 8x_2 = 4 \\ 9x_1 & + & 2x_2 = -2 \end{array}$$



Row-Echelon and Reduced Row-Echelon

A matrix (any matrix , not just an augmented matrix) is said to be in reduced row - echelon form if it satisfies all four of the following conditions.

- 1 If there are any rows of all zeros then they are at the bottom of the matrix
- 2 If a row does not consist of all zeros then its first non - zero entry (i.e the left most non - zero entry) is a 1. This 1 is called a leading 1.



Cont...

- ③ In any two successive rows, neither of which consists of all zeroes, the leading 1 of the lower row is to the right of the leading 1 of the higher row.
- ④ If a column contains a leading 1, then all the other entries of the column are zero.



Cont...

A matrix (again any matrix) is said to be in row - echelon form if it satisfies items 1 – 3 of the reduced row - echelon form definition.

Notice from these definitions that a matrix that is in reduced row - echelon form is also in row echelon form while a matrix in row - echelon form may or may not be in reduced row-echelon form.



Example 48

The following matrices are all in row-echelon form.

$$\textcircled{1} \begin{bmatrix} 1 & 0 & \underline{5} \\ 0 & 1 & \underline{3} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\textcircled{2} \begin{bmatrix} 1 & \underline{-8} & 10 & \underline{5} & -3 \\ 0 & 1 & 13 & \underline{9} & 12 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

None of the matrices in the previous example are in reduced-echelon form. The entries are preventing these matrices from being in the reduced row-echelon form; all these underlined entries would need to be zero.



EXAMPLE

The following matrices are all in row-echelon form.

$$(a) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$(b) \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(c) \begin{bmatrix} 0 & 1 & 0 & -8 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(d) \begin{bmatrix} 1 & -7 & 10 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



Solving Systems Of Equations-Example

Question

Use the Gaussian Elimination and Gauss - Jordan Elimination method to solve the following system of linear equations.

$$-2x_1 + x_2 - x_3 = 4$$

$$x_1 + 2x_2 + 3x_3 = 13$$

$$3x_1 + x_3 = -1$$

Cont...

Solution

Getting the augmented matrix for this system.

$$\left[\begin{array}{cccc} \underline{-2} & 1 & -1 & 4 \\ 1 & 2 & 3 & 13 \\ 3 & 0 & 1 & -1 \end{array} \right]$$



Cont...

Solution

Interchanging Row 1 and row 2, we have

$$\begin{bmatrix} \underline{-2} & 1 & -1 & 4 \\ 1 & 2 & 3 & 13 \\ 3 & 0 & 1 & -1 \end{bmatrix} R_1 \leftrightarrow R_2 \begin{bmatrix} 1 & 2 & 3 & 13 \\ \underline{-2} & 1 & -1 & 4 \\ \underline{3} & 0 & 1 & -1 \end{bmatrix}$$

Next

$$\begin{bmatrix} 1 & 2 & 3 & 13 \\ \underline{-2} & 1 & -1 & 4 \\ \underline{3} & 0 & 1 & -1 \end{bmatrix} R_2 + 2R_1, R_3 - 3R_1 \rightarrow \begin{bmatrix} 1 & 2 & 3 & 13 \\ 0 & \underline{5} & 5 & 30 \\ 0 & -6 & -8 & -40 \end{bmatrix}$$

Cont...

Solution

Dividing the second row by 5 since that don't give as fractions. We have

$$\begin{bmatrix} 1 & 2 & 3 & 13 \\ 0 & \underline{5} & 5 & 30 \\ 0 & -6 & -8 & -40 \end{bmatrix} \xrightarrow{\frac{1}{5}R_2} \begin{bmatrix} 1 & 2 & 3 & 13 \\ 0 & 1 & 1 & 6 \\ 0 & \underline{-6} & -8 & -40 \end{bmatrix}$$
$$\begin{bmatrix} 1 & 2 & 3 & 13 \\ 0 & 1 & 1 & 6 \\ 0 & \underline{-6} & -8 & -40 \end{bmatrix} \xrightarrow{R_3 + 6R_2} \begin{bmatrix} 1 & 2 & 3 & 13 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & \underline{-2} & -4 \end{bmatrix}$$

Cont...

Solution

Multiplying row 3 by $-\frac{1}{2}$, we have

$$\begin{bmatrix} 1 & 2 & 3 & 13 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & \underline{-2} & -4 \end{bmatrix} -\frac{1}{2}R_2 \rightarrow \begin{bmatrix} 1 & 2 & 3 & 13 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$



Cont...

Solution

At this point the augmented matrix is in row-echelon form. Going back to equations gives,

$$\begin{bmatrix} 1 & 2 & 3 & 13 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

$$x_1 + 2x_2 + 3x_3 = 13, \quad x_2 + x_3 = 6, \quad x_3 = 2$$

Cont

Solution

Summarizing up the solution to the system is,

$$x_1 = -1, x_2 = 4, x_3 = 2$$

Now picking back up at the row - echelon form of the matrix and further reduce the matrix into reduced row-echelon form.



Cont...

Solution

we have

$$\begin{bmatrix} 1 & 2 & \underline{3} & 13 \\ 0 & 1 & \underline{1} & 6 \\ 0 & 0 & 1 & 2 \end{bmatrix} R_1 - 3R_3, R_2 - R_3 \rightarrow \begin{bmatrix} 1 & \underline{2} & 0 & 7 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Finally

$$\begin{bmatrix} 1 & \underline{2} & 0 & 7 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 2 \end{bmatrix} R_1 - 3R_3 \rightarrow \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Solution

Now using the Gauss- Jordan Elimination gives the equations.

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

$$\therefore x_1 = -1, x_2 = 4, x_3 = 2$$



Exercise 7

Solve the following system of linear equations.

$$x_1 - x_2 + x_3 = -2$$

$$-x_1 + x_2 - 2x_3 = 3$$

$$2x_1 - x_2 + 3x_3 = 1$$



Homogeneous Systems Of Linear equations

A system of n linear equations in m unknowns in the form

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1m}x_m = 0$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2m}x_m = 0$$

$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nm}x_m = 0$$

is called homogeneous system.



Cont...

The one characteristic that defines a homogeneous system is the fact that all the equations are set to zero. Hopefully, it is clear that if we take

$$x_1 = 0, x_2 = 0, \dots, x_m = 0$$

we will have a solution to the homogeneous system of equations.



Cont...

In other words, with a homogeneous system, we are guaranteed to have at least one solution .

This means that Theorem 1 from the previous section can then be reduced to the following for homogeneous systems.



Theorem 2

Given a homogeneous system of n equations and m unknowns there will be one of two possibilities for solutions to the system.

- 1 There will be exactly one solution , $x_1 = 0, x_2 = 0, \dots x_m = 0$. This solution is called the trivial solution.
- 2 There will be infinitely many non-zero solutions in addition to the trivial solution.

When we say non-zero solution, It is completely possible that some of them will still be zero, but at least one will



THEOREM 3

Given a homogeneous system of n linear equations in m unknowns, if $m > n$ (i.e there are more unknowns than equations) there will be infinitely many solutions to the system.



The Rank of a Matrix

The **rank** of a matrix A , denoted $\text{rank}(A)$, is the maximum number of **linearly independent** rows or columns in the matrix. If a vector is **linearly independent** of a set of other vectors, that means it cannot be written as a **linear combination** of them.



Theorem 4

If A is any matrix, then $\text{rank}(A) = \text{rank}(A^T)$.



Theorem 5

If A is an $m \times n$ matrix , then:

- (a) $rank(A)$ = the number of leading variable in the solution of $Ax = 0$.
- (b) $nullity(A)$ = the number of parameters in the solution $Ax = 0$.



Theorem 6

If A is a matrix with n columns then ,
$$n = \text{rank}(A) + \text{nullity}(A)$$



Example 49

Question

Find the rank and nullity of

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 6 & -3 & -1 \\ 3 & 10 & -6 & -5 \end{bmatrix}$$



Cont...

Solution

By using elementary row operations, we transform the matrix A as follows:

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 6 & -3 & -1 \\ 3 & 10 & -6 & -5 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 2 & -3 & -1 \\ 0 & 4 & -6 & -2 \end{bmatrix}$$



Cont...

Solution

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 2 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Therefore,

$\text{Rank}(A) = 2$ and $\text{nullity}(A) = 4 - 2 = 2$.



Exercise 8

Find the rank and nullity of

$$A = \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix}$$



The Consistent Theorem

If $Ax = b$ is a linear system of m equations in n unknowns, Then the following are equivalent.

- (a) $Ax = b$ is consistent
- (b) The coefficient matrix A and the augmented matrix $[A|b]$ have the same rank. In other words, a linear system is said to be consistent if and only if the rank of A is equal to the augmented matrix $[A|b]$.



Cont...

(c) The system has a solution if and only if $\text{rank}(A) = \text{rank}[A|b]$.

If the rank is equal to n , then the system has a unique solution. If $\text{rank}(A) = \text{rank}[A|b]$ but the $\text{rank} < n$, there are an infinite number of solutions. If we denote the rank by r , then r of the unknown variables can be expressed as linear combinations of $n - r$ of the other variables.



Example 50

Question

Find k such that the following system is consistent

$$\begin{aligned}x - 2y - z &= 1 \\ -x - 3y + z &= -1 \\ -5x - 8y + 5z &= k\end{aligned}$$



Cont...

Solution

We first put the above systems in the augmented form.
Thus

$$\left[\begin{array}{cccc} 1 & 2 & -1 & 1 \\ -1 & -3 & 1 & -1 \\ -5 & -5 & 5 & k \end{array} \right]$$



Cont...

Solution

Applying row operations and reducing the augmented matrix to echelon form. We have

$$\begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & k+3 \end{bmatrix}$$

The rank of A is 2 and the rank of $[A|B] = 0$.



Cont...

Solution

This is possible if $k + 3 = 0 \therefore k = -3$.

$$x - 2y - z = 1$$

$$-x - 3y + z = -1$$

$$-5x - 8y + 5z = k$$

is consistent if and only if $k = -3$.



Example 51

Question

Show that

$$x_1 - 2x_2 - 3x_3 + 2x_4 = -4$$

$$-3x_1 + 7x_2 - x_3 + x_4 = -3$$

$$2x_1 - 5x_2 + 4x_3 - 3x_4 = 7$$

$$-3x_1 + 6x_2 + 9x_3 - 6x_4 = -1$$

is inconsistent.

Cont...

Solution

In augmented form:

$$\left[\begin{array}{cccc|c} 1 & -2 & -3 & 2 & -4 \\ -3 & 7 & -1 & 1 & -3 \\ 2 & -5 & 4 & -3 & 7 \\ -3 & 6 & 9 & -6 & -1 \end{array} \right]$$



Cont...

Solution

which has a reduced row echelon form.

$$\left[\begin{array}{cccc|c} 1 & 0 & -23 & 16 & 0 \\ 0 & 1 & -10 & 7 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

the system is inconsistent because $\text{rank}(A) \neq \text{rank}[A|b]$

Example 52

Question

Find the general solution

$$4x + 3y + 2z = 0$$

$$-x + 2y + 2z + w = 0$$

$$3x + 5y + 4z + w = 0$$



Cont...

Solution

The coefficient matrix is

$$\begin{bmatrix} 4 & 3 & 2 & 0 \\ -1 & 2 & 2 & 1 \\ 3 & 5 & 4 & 1 \end{bmatrix}$$



Cont...

Solution

$R_2 = R_2 - 4R_1$, $R_3 = R_3 - 3R_1$ which gives

$$\begin{bmatrix} 1 & -2 & -2 & -1 \\ 0 & 11 & 10 & 4 \\ 0 & 11 & 10 & 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -2 & -2 & -1 \\ 0 & 11 & 10 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



Cont...

Solution

$$\begin{bmatrix} 1 & -2 & -2 & -1 \\ 0 & 1 & \frac{10}{11} & \frac{4}{11} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The above is equivalent to

$$x - 2y - 2z - w = 0$$

$$y + \frac{10}{11}z + \frac{4}{11}w = 0$$

Cont...

Solution

Let $z = 11s$ and $w = 11t$ where s and t are arbitrary real numbers then we have

$$y = -10s - 4t$$

$$x - 2y = 22s + 11t$$

$$x = -20s - 8t + 22s + 11t = 2s + 3t$$



Cont...

solution

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = s \begin{bmatrix} 2 \\ -10 \\ 11 \\ 0 \end{bmatrix} + t \begin{bmatrix} 3 \\ -4 \\ 0 \\ 11 \end{bmatrix}$$



Eigenvalues And Eigenvectors

Definition 1

Suppose that A is an $n \times n$ matrix .

Also suppose that x is a non-zero vector from $\mathbb{R}^n$ and that λ is any scalar (this can be zero) so that

$$Ax = \lambda x \quad (107)$$

We then call x an eigenvector of A and λ an eigenvalue of A .



Finding Eigenvalues

Suppose that $A = \begin{bmatrix} 6 & 16 \\ -1 & -4 \end{bmatrix}$ then $x = \begin{bmatrix} -8 \\ 1 \end{bmatrix}$ is an eigenvector with corresponding eigenvalue 4 because

$$Ax = \begin{bmatrix} 6 & 16 \\ -1 & -4 \end{bmatrix} \begin{bmatrix} -8 \\ 1 \end{bmatrix} = \begin{bmatrix} -32 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} -8 \\ 1 \end{bmatrix} = \lambda x$$



Cont...

Let's start with $Ax = \lambda x$ and rewrite it as follows,

$$Ax = \lambda Ix \quad (108)$$

Note that all we did was to insert the identity matrix into the right side.

It follows that

$$Ax - \lambda Ix = 0 \quad (109)$$



Cont...

and

$$(A - \lambda I)x = 0 \quad (110)$$

This system will have a non-zero solution if and only if

$$\det(A - \lambda I) = 0 \quad (111)$$

So, eigenvalues will be scalars , λ for which the matrix $A - \lambda I$ will be singular, i.e $\det(A - \lambda I) = 0$.



Definition 2

Suppose A is an $n \times n$ matrix then, $\det(A - \lambda I) = 0$ is called the characteristic equation of A .

$$p(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0 \quad (112)$$

called the characteristic polynomial of A .



Definition 3

Suppose A is an $n \times n$ matrix and that $\lambda_{1,2,\dots,n}$ is the complete list of all eigenvalues of A including repeats. If λ occurs exactly once in this list then we call λ a simple eigenvalue. If λ occurs $k \geq 2$ times in the list we say that λ has a multiplicity of k .



Example 53

Question

Find all the eigenvalues for the given matrix

$$A = \begin{bmatrix} 6 & 16 \\ -1 & -4 \end{bmatrix}$$



Cont...

Solution

First we need the matrix $A - \lambda I$.

$$A - \lambda I = \begin{bmatrix} 6 & 16 \\ -1 & -4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 6 - \lambda & 16 \\ -1 & -4 - \lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (6 - \lambda)(-4 - \lambda) + 16 = \lambda^2 - 2\lambda - 8$$

Now , set this equation to zero and solve for the eigenvalues

Cont...

Solution

$$\lambda^2 - 2\lambda - 8 = (\lambda - 4)(\lambda + 2) \implies \lambda_1 = -2, \lambda_2 = 4$$

The eigenvalues are both simple eigenvalues.



3×3 Matrices

If $A = [a_{ij}]$ is a 3×3 , then the calculation of $\det(A - \lambda I)$ can be done directly , or the following formulas for the coefficients can be used . By collecting terms we can put the determinant of $(A - \lambda I)$ into the form

$$\det(A - \lambda I) = -\lambda^3 + \operatorname{tr}(A)\lambda^2 - pm(A)\lambda + \det(A)$$



Cont...

Where the coefficients of λ^2 and λ are as follows:

$$tr(A) = a_{11} + a_{22} + a_{33}$$

is the sum of the diagonal entries, called the *trace* of A ,

and $pm(A) = M_{11} + M_{22} + M_{33} =$

$$\det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} + \det \begin{bmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{bmatrix} + \det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$
 is the

sum of the minors of the main diagonal entries, called the principal minors of A



Example 54

Question

Find the eigenvalues of the matrix A

$$A = \begin{bmatrix} 3 & -1 & 2 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$



Solution

The characteristic equation of A is $\det(A - \lambda I)$, thus

$$\det \begin{bmatrix} 3 - \lambda & -1 & 2 \\ -1 & 1 - \lambda & 0 \\ 2 & 0 & 1 - \lambda \end{bmatrix} = -\lambda^3 + 5\lambda^2 - 2\lambda - 2 = 0$$

Multiplying both sides of the equation by (-1) , we have ,
 $\lambda^3 - 5\lambda^2 + 2\lambda + 2 = 0$



Cont...

Solution

We can then find that $\lambda = 1$ is a root. And the factored form is

$$(\lambda - 1)(\lambda^2 - 4\lambda - 2) = 0$$

Using the quadratic formula on the second factor, we get $\lambda = 2 \pm \sqrt{6}$. Thus the eigenvalues of A are $\lambda_1 = 1$, $\lambda_2 = 2 + \sqrt{6}$, $\lambda_3 = 2 - \sqrt{6}$.

Finding Eigenvectors

If λ is an eigenvalue of a matrix A , then its associated eigenvectors are the non-zero solutions of the homogeneous system of equations.

$$(A - \lambda I)x = 0$$



Example 55

Question

The Eigenvalues of the matrix

$$A = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$$

are $\lambda_1 = 5$ and $\lambda_2 = -2$. Find the associated eigenvectors for each eigenvalue.



Cont...

Solution

First we take $\lambda_1 = 5$ which gives us the homogeneous system

$$(A - \lambda I)x = \begin{bmatrix} 1-5 & 4 \\ 3 & 2-5 \end{bmatrix} x = \begin{bmatrix} -4 & 4 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



Cont...

Solution

The solution space consist of all vectors $x = (x_1, x_2)$ such that

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



Cont...

Solution

When $\lambda_1 = -2$ the solution space consist of all vectors $x = (x_1, x_2)$ such that

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = t \begin{bmatrix} -4 \\ 3 \end{bmatrix}$$



Example on 3×3 matrices

Question

Find the Eigenvalues and the corresponding Eigenvectors of the matrix

$$A = \begin{bmatrix} 3 & -2 & -2 \\ -2 & 3 & 2 \\ 3 & -3 & -2 \end{bmatrix}$$



Cont...

Solution

We calculate the determinant

$$\det(A - \lambda I) = \begin{vmatrix} 3 - \lambda & -2 & -2 \\ -2 & 3 - \lambda & 2 \\ 3 & -3 & -2 - \lambda \end{vmatrix} = -\lambda^3 + 4\lambda^2 - 5\lambda + 2 = 0$$

Then multiply by -1 , we get cubic equation

$$\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$$

Cont...

Solution

We find that $\lambda_1 = 1$ and $\lambda_2 = 2$ by testing the numbers $\pm 2, \pm 1$ for roots.

We obtain the factored form

$$(\lambda - 1)^2(\lambda - 2) = 0$$

Thus we have the eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$, where $\lambda_1 = 1$ has a multiplicity two(2)

Cont...

Solution

(a) For $\lambda_1 = 1$
we get the system

$$(A - 1I)x = \begin{bmatrix} 2 & -2 & -2 \\ -2 & 2 & 2 \\ 3 & -3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

the coefficient matrix has the echelon form



Cont...

Solution

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Therefore we may set $x_3 = t$, $x_2 = s$ and get $x_1 = s + t$.



Solution

Thus ,

$$x = s \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$



Cont...

Solution

(b) For $\lambda_2 = 2$

we get the system

$$(A - 2I)x = \begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & 2 \\ 3 & -3 & -4 \end{bmatrix} x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Cont...

Solution

Verify as an exercise that the solutions are all vectors of the form

$$x = t \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}$$



Thank You for Coming

